

EXP 8

Consider a Online College Notice Portal. Develop a JavaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration). Note: Your application must access a database using Servlet/JSP.

AIM:

To develop an Online Quiz Application that validates user inputs using JavaScript (client-side validation) and stores quiz/user data in a database using Servlet/JSP (server-side processing).

ALGORITHM:

- Start the application.
- Display the quiz registration/login form.
- Accept user inputs such as:
 - Name
 - Email
 - Password
 - Quiz category
- Perform client-side validation using JavaScript:
 - Check empty fields
 - Validate email format
 - Validate password strength
 - If validation fails:
 - Show error messages
 - Prevent form submission
 - If validation succeeds:
 - Submit form to Servlet
- On server side:
 - Receive data
 - Perform server-side validation
 - Connect to database using JDBC
 - Store user details in database
 - Allow user to attempt quiz
 - Display confirmation/result
- Stop the application

STEPS:

1. Design the registration form with fields like name, email, password, phone number, exam selection, and date.
2. Implement JavaScript validation for all form controls.
3. Ensure proper error messages are displayed for invalid inputs.
4. Configure Servlet/JSP to handle form submission.
5. Establish database connectivity using JDBC.
6. Create a database table for storing registration details.
7. Insert validated data into the database.
8. Retrieve and display confirmation details to the user.
9. Handle duplicate registrations and invalid data cases.
10. Test the application for different input scenarios.

Program:

```
<!DOCTYPE html>
<html>
<head>
<title>Online Quiz Registration</title>

<script>
function validateForm() {
let name = document.forms["quizForm"]["name"].value;
let email = document.forms["quizForm"]["email"].value;
let password = document.forms["quizForm"]["password"].value;

let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

if (name == "" || email == "" || password == "") {
alert("All fields are required");
return false;
}

if (!email.match(emailPattern)) {
alert("Invalid email format");
return false;
}
```

```

        if (password.length < 6) {
            alert("Password must be at least 6 characters");
            return false;
        }

        return true;
    }
</script>
</head>

<body style="font-family: Segoe UI;text-align:center;margin-top:100px;">

<h2>Online Quiz Registration</h2>

<form name="quizForm" action="RegisterServlet" method="post" onsubmit="return validateForm()">

    <input type="text" name="name" placeholder="Name"><br><br>
    <input type="text" name="email" placeholder="Email"><br><br>
    <input type="password" name="password" placeholder="Password"><br><br>

    Category:
    <select name="category">
        <option>Java</option>
        <option>HTML</option>
    </select><br><br>

    <button type="submit">Register & Start Quiz</button>

</form>

</body>
</html>

```

RegisterServlet.java (Database Insert)

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
import java.sql.*;

@WebServlet("/RegisterServlet")
public class RegisterServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String name = request.getParameter("name");
        String email = request.getParameter("email");
        String password = request.getParameter("password");
        String category = request.getParameter("category");

        response.setContentType("text/html");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/quiz_db",
                "root",
                "root"
            );

            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO users(name, email, password, category) VALUES (?, ?, ?, ?)"
            );

            ps.setString(1, name);
            ps.setString(2, email);
            ps.setString(3, password);
            ps.setString(4, category);

            int i = ps.executeUpdate();

            if (i > 0) {
                response.sendRedirect("quiz.jsp");
            } else {
                response.getWriter().println("Registration Failed");
            }

            ps.close();
            con.close();

        } catch (Exception e) {
            e.printStackTrace();
            response.getWriter().println("Error: " + e.getMessage());
        }
    }
}

```

quiz.jsp (Simple Quiz Page)

```
<%@ page language="java" %>
<html>
<body style="font-family: Segoe UI;">

<h2>Online Quiz</h2>

<form action="ResultServlet" method="post">

<p>1. Java is a?</p>
<input type="radio" name="q1" value="a"> Language
<input type="radio" name="q1" value="b"> OS

<br><br>

<button type="submit">Submit Quiz</button>

</form>

</body>
</html>
```

ResultServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;

@WebServlet("/ResultServlet")
public class ResultServlet extends HttpServlet {

    protected void doPost(HttpServletRequest req, HttpServletResponse res)
        throws IOException {

        String q1 = req.getParameter("q1");

        int score = 0;

        if ("a".equals(q1)) score++;

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        out.println("<h2>Your Score: " + score + "/1</h2>");
        out.println("<a href='index.html'>Back</a>");
    }
}
```

OUTPUT :

Create Quiz

Title *

General Knowledge Quiz

Description *

A quiz on general knowledge question and answers.

Total Questions *

10

Marks per Question *

1

Category *

General Knowledge

Create Quiz

All Quizzes

#	Quiz Title	Description	Questions	Marks	Category	Created On	Action
1	General Knowledge Quiz	A quiz on general knowledge question and answers.	10	10	General Knowledge	2026-04-22	View Edit
2	Java Basics Quiz	Test your knowledge on Java basics.	15	15	Programming	2026-05-01	View Edit
3	Web Development Quiz	Questions on HTML, CSS, JavaScript.	12	12	Web Development	2026-05-05	View Edit
4	Database Quiz	MCQ questions on DBMS concepts.	10	10	Database	2026-05-10	View Edit
5	Aptitude Quiz	Aptitude and reasoning questions.	20	20	Aptitude	2026-05-12	View Edit

RESULT :

The Online Quiz Application successfully validates user inputs using JavaScript, processes data using Servlet/JSP, and stores user details in a database, ensuring proper validation and dynamic quiz functionality.